Task 2 Report

Multiple Sequence Alignment (MSA) of BRCA1 Protein Sequences

OBJECTIVE

To perform Multiple Sequence Alignment (MSA) of five homologous BRCA1 protein sequences using Clustal Omega and identify conserved regions and motifs among the selected species.

PROTEIN FAMILY

Breast Cancer Type 1 Susceptibility Protein (BRCA1)

BRCA1 is a conserved protein involved in important cellular processes including DNA damage response and DNA repair. The selected sequences belong to the same protein family and have a length of approximately 1863 amino acids.

SEQUENCES USED

Species/Organism	Protein Accession	Protein Length
Homo sapiens	P38398	1863 aa
Pan troglodytes	Q9GKK8.2	1863 aa
Gorilla gorilla gorilla	Q6J6I8.1	1863 aa
Pongo abelii	XP_003778880.3	1863 aa
Additional BRCA1 homolog	NP_001108421.1	~1863 aa

TOOL USED

Clustal Omega was used to perform the Multiple Sequence Alignment.

The five BRCA1 protein sequences were submitted to Clustal Omega in FASTA format. The resulting alignment was examined to identify conserved amino acid positions and regions.

MSA OVERVIEW

The complete alignment shows a high degree of sequence similarity among the five BRCA1 protein sequences. Many amino acid positions are conserved throughout the alignment, demonstrating that the BRCA1 protein is strongly conserved among the selected organisms.

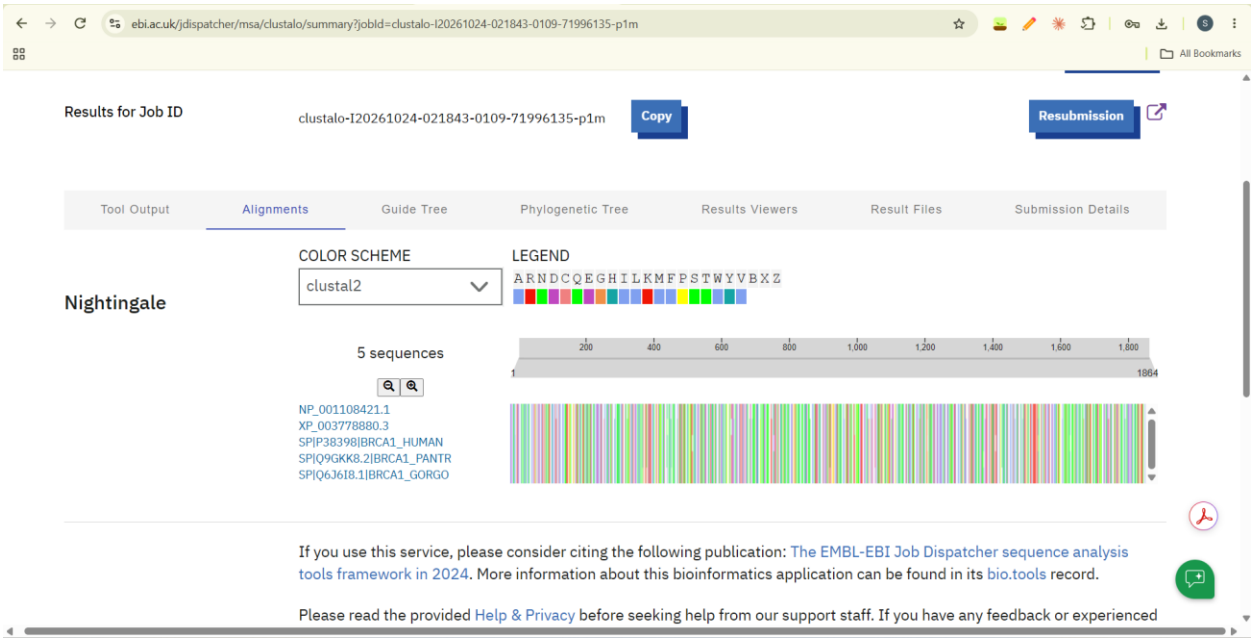


Figure 1. Multiple Sequence Alignment overview of the five BRCA1 protein sequences generated using Clustal Omega.

CONSERVED REGIONS

1. N-terminal Conserved Region

The N-terminal region shows strong conservation among the five sequences, including conserved cysteine and histidine residues associated with the BRCA1 RING domain.



Figure 2. Conserved N-terminal region of the BRCA1 protein sequences.

2. Conserved Region 1

This region contains highly conserved amino acids among the five BRCA1 sequences, indicating an important functional role.

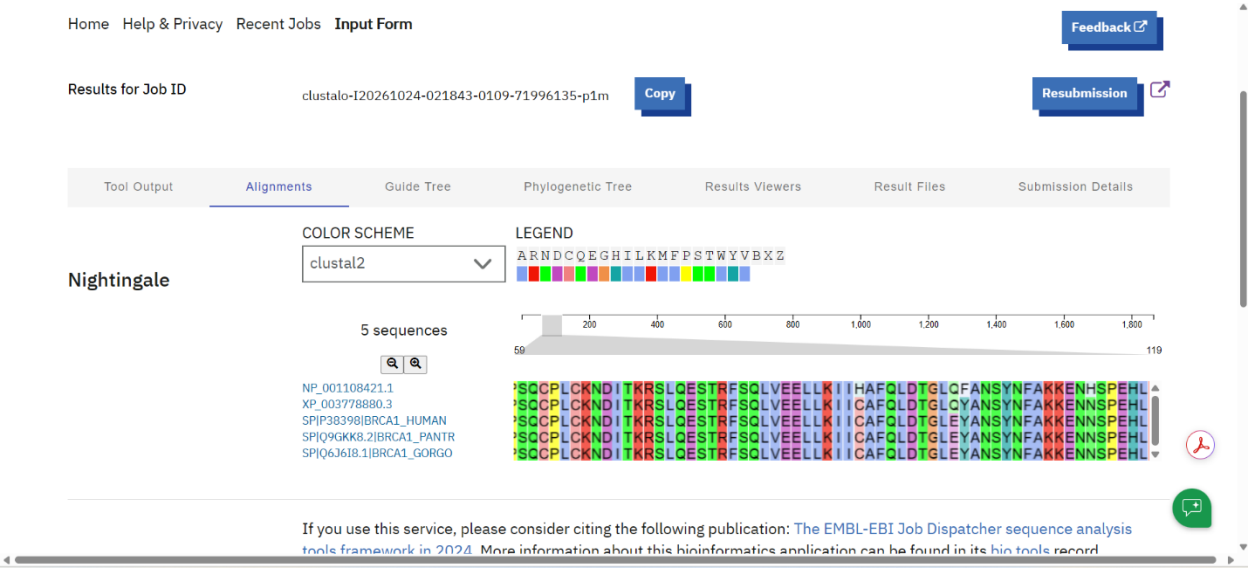


Figure 3. Conserved region 1 identified in the MSA.

3. Conserved Region 2

This region shows strong conservation across the five sequences, suggesting structural and functional importance.

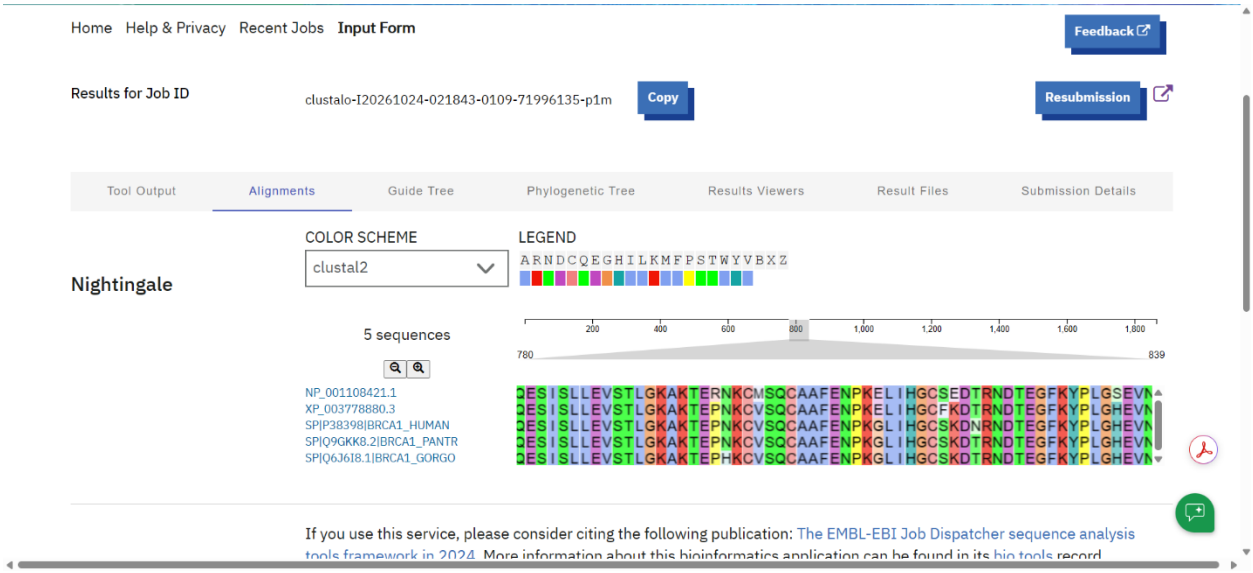


Figure 4. Conserved region 2 identified in the MSA.

4. Central/Cysteine-rich Conservation (Positions 600–960)

This region shows strong conservation, particularly around cysteine-rich residues, suggesting an important structural and functional role.



Figure 5. Conserved central/cysteine-rich region of BRCA1.

5. C-terminal Conservation (Positions 1600–1863)

The C-terminal region is highly conserved among the five sequences, indicating strong evolutionary and functional importance.

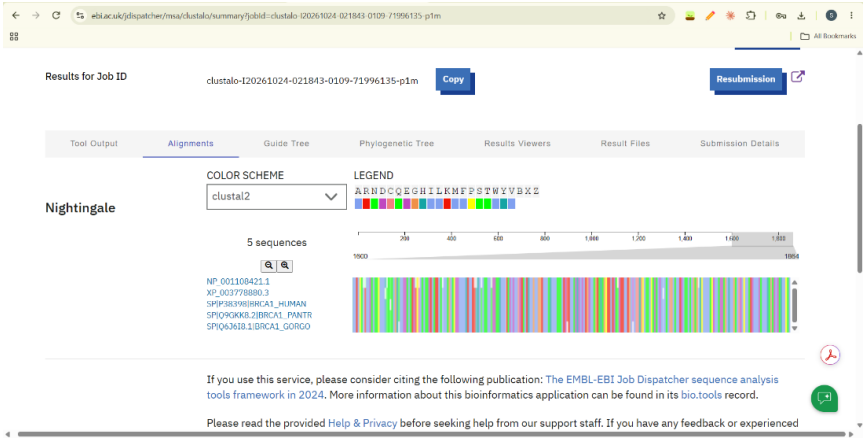


Figure 6. Conserved C-terminal region of BRCA1.

INTERPRETATION

The Multiple Sequence Alignment demonstrates substantial conservation among the five BRCA1 protein sequences. The high similarity is particularly evident in the N-terminal, central conserved/cysteine-rich, and C-terminal regions.

Conserved amino acid residues are likely important for maintaining the structure and biological function of BRCA1. The presence of conserved regions across different species also indicates evolutionary conservation of the BRCA1 protein.

Overall, the MSA supports that the selected sequences are homologous BRCA1 proteins belonging to the same protein family.

CONCLUSION

Clustal Omega successfully aligned the five BRCA1 protein sequences. The alignment revealed several highly conserved regions, including the N-terminal region, central/cysteine-rich region, and C-terminal region. These conserved regions indicate that important structural and functional features of BRCA1 have been maintained during evolution.

SOFTWARE/DATABASE

- **NCBI Protein Database** – used to obtain protein sequences and accession information.
- **Clustal Omega** – used for Multiple Sequence Alignment.
- **FASTA format** – used for sequence submission.